

Using a Sandboxed Ticketing Bot to Teach Digital Trust and IT Audit*

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TO DO: - create video - sharpen instructions on Github - wait for IRB approval on the project as secondary data - update team after meeting with lorraine lee - send invite to lorraine (overleaf) - create manual - change structure of paper - Issues

Abstract

The verification of Information Technology (IT) controls is a core responsibility of IT auditors. This case places students in the role of IT auditors assigned to assess the effectiveness of controls over a sandboxed online ticketing system. Students first participate in limited ticket drops, inspect the website's Document Object Model (DOM), and use AI-assisted coding to understand how automated purchasing can occur. Students then evaluate whether the system's controls over purchase limits, transaction authorization, backend logging, exception reporting, and monitoring are effective. The case introduces students to technical skills such as browser inspection, basic automation logic, and AI-assisted coding while maintaining an internal-control focus. Students document their work through a DOM discovery worksheet, ticket-drop observations, an internal-control review matrix, and a short reflection. Digital trust, automated transactions, IT controls, and AI-assisted technical fluency are all topics relevant to Accounting Information Systems (AIS), IT Audit, and Audit courses. As such, instructors can use this case at the undergraduate or graduate level in an Accounting Information Systems, IT Audit, or Audit class.

Keywords: IT General Controls, Digital Trust, Ticketing Bots, Internal Controls, AI-Assisted Coding, Accounting Information Systems

1 Introduction

The Sarbanes-Oxley Act of 2002 (SOX) requires management to maintain and assess internal controls over financial reporting. Under PCAOB AS 2201, auditors evaluate these controls using a top-down approach that begins with financial statement risks and then focuses on the controls most relevant to material misstatement. Information Technology (IT) general controls are central to this process because they support the reliability of application controls, transaction processing, and system-generated data.

Prior AIS education cases have emphasized user access management, database design, system implementation, and other traditional IT control topics (Lee and Sawyer, 2019; Mascha and Miller, 2024; Jiang, Luttenton-Knoll, Hillman, and Pellathy, 2024a). These topics remain important, but contemporary IT auditors must also understand automated transactions, platform governance, data analytics, cybersecurity controls, artificial intelligence, and the risks created by rapidly changing digital systems (Dzuranin and Mălăescu, 2016; Li and Goel, 2025). As business processes become increasingly electronic, auditors must understand how transactions are initiated, authorized, processed, recorded, and monitored within digital platforms.

Technical fluency is therefore becoming part of audit competence. Coding is increasingly relevant to accountants because it allows professionals to automate repetitive tasks, inspect data and systems directly, and understand the logic embedded in digital business processes (ACCA, 2021). Recent professional commentary similarly notes that accountants are already using AI for tasks such as automation, data analysis, technical research, and document preparation (Kenney, 2025). These developments suggest that auditors need a certain level of coding and AI literacy to evaluate system behavior, validate automated outputs, and communicate effectively with technical specialists.

A challenge for AIS instructors is to teach technical skills without turning the course into a programming course. The instructional setting must be simple enough for students to understand quickly, but rich enough to demonstrate how system design affects internal controls. We use online ticketing as this setting. Students understand the business process, the scarcity of tickets creates an intuitive incentive for automation, and the transaction flow allows instructors to connect technical behavior to audit concepts. The *Wall Street Journal* has reported on ticket scalping, resale practices, and ticketing fees, reflecting a broader phenomenon that students are likely to recognize (Steele, 2025; Fanelli and Michaels, 2026). Using a real-life example gives the case verisimilitude and engages students, while allowing instructors to keep the implementation focused on coding, digital transaction processes, and internal controls.

This paper presents an instructional case that uses a sandboxed ticketing website to teach IT audit and internal-control concepts through a controlled ticket-drop exercise. Students participate in three limited ticket releases, inspect the website’s HTML structure and Document Object Model (DOM), and use AI-assisted “vibe coding” to improve their ability to purchase one ticket for each event. The instructor then reveals the backend logs and the automation script, allowing students to compare front-end behavior with system-recorded activity. Students subsequently work in groups to identify control weaknesses, evaluate evidence from the exercise, and recommend preventive and detective controls. The exercise uses gamification Blix, Edmonds, Edmonds, and Sorensen (2026) and a real-world ticket-scalping context (Gantman, 2024) to help students connect coding, digital transaction processes, backend logging, and internal-control

reasoning.

The purpose of the case is not to teach students how to scalp tickets. Rather, the case teaches students how automated behavior exposes weaknesses in digital internal controls. By requiring students to inspect the DOM, observe automated transactions, test control assertions, analyze exception reports, and prepare audit documentation, the case responds to calls for AIS education to integrate technical fluency, AI-related skills, and audit judgment (AICPA, 2025; Dzuraniin and Mălăescu, 2016; Li and Goel, 2025; ACCA, 2021; Kenney, 2025).

1.1 Real-Life Background

In September 2025, the Federal Trade Commission, joined by seven states, sued Live Nation and Ticketmaster for allegedly engaging in illegal ticket resale practices and deceiving artists and consumers about prices and ticket limits. The FTC alleged that Ticketmaster publicly claimed to protect fans from brokers while benefiting from brokers who used thousands of accounts and proxy IP addresses to bypass stated ticket limits, acquire large volumes of tickets, and resell them on Ticketmaster's own resale platform at substantial markups (Federal Trade Commission, 2025).

These allegations provide a useful setting for teaching accounting information systems because they show that ticketing failures are not only consumer-protection or cybersecurity problems. They are also internal-control problems. A ticketing platform may record a sale correctly, but the transaction may still violate purchase limits, platform rules, artist restrictions, or anti-bot requirements. In that case, the data may be mechanically complete but economically misleading.

From an IT audit perspective, the allegations raise direct control questions. Were purchase limits actually enforced? Did the company monitor high-volume brokers, repeated purchases, proxy activity, or unusual account behavior? Were exceptions reviewed and remediated? Did resale revenues depend on activity that violated the company's stated rules? These questions connect bot activity to core audit concepts, including authorization, transaction validity, monitoring, access control, and management override.

The FTC also alleged that Ticketmaster advertised prices that excluded mandatory fees and markups, with some fees reaching 44 percent of the ticket cost and total fees from 2019 to 2024 reaching \$16.4 billion (Federal Trade Commission, 2025). This allegation adds a second audit-relevant issue: whether the system presents complete and accurate transaction information before purchase.

This case translates those issues into a sandboxed classroom ticketing website. Students first experience a limited ticket release as ordinary users. They then inspect the website's HTML structure and DOM elements, use AI-assisted coding to understand automated interaction, and test whether the platform's controls prevent, detect, or correct bot-like purchasing behavior. The purpose is not to train students to scalp tickets, but to show that IT auditors must understand how digital transactions are initiated, authorized, processed, logged, monitored, and reported.

2 Literature Review

Following the structure of [Lee and Sawyer \(2019\)](#), we review recent instructional cases, in the last five years, related to internal controls and IT controls from three leading accounting education journals: *Issues in Accounting Education*, *Journal of Accounting Education*, and *AIS Educator Journal*. Table 1 summarizes the cases most closely related to this study.

Table 1: Recent instructional cases on internal controls and IT controls (2021–2026)

Year	Citation	Full title	Primary topics & software	Main task related to IT control
2021	Lawson and Street (2021)	Detecting dirty data using SQL: Rigorous house insurance case	Data quality, claims analytics, preventative IS controls; software: SQL	Query claims data, identify integrity exceptions, and recommend preventative information-system controls.
2022	Boss et al. (2022)	Accountants, Cybersecurity Isn't Just for "Techies": Incorporating Cybersecurity into the Accounting Curriculum	Cybersecurity, disclosure, audit/tax risks, short cases; artefacts: case materials and 10-K-style disclosures	Analyse six cyber scenarios and evaluate risk, control, reporting, and assurance implications.
2022	Brown and Lightle (2022)	Only Reliable Parts and Supplies, Inc.: Assessing and documenting the design of internal controls	Sales-process controls, flowcharting, audit documentation; software: flowcharting/word-processing tools	Flowchart the sales cycle, assess control design, and draft a memo on internal-control weaknesses.
2023	Parlier and Lee (2023)	Inventory analytics: A teaching case using excel and Alteryx	Audit analytics, inventory anomalies, exception analysis; software: Excel and Alteryx	Use analytics to investigate inventory anomalies and discuss implications for control monitoring and audit evidence.
2023	Hawk (2023)	Building Higher-Order Thinking Skills Through Assessment of User Account Access	IT general controls, user access management, audit analytics; software: Excel and CaseWare IDEA	Test user-access controls, analyse exceptions, and conclude on ITGC effectiveness.
2024	Boss et al. (2024)	Be an Expert: A Critical Thinking Approach to Responding to High-Profile Cybersecurity Breaches	Cyber breaches, internal audit, incident response; software: case materials	Evaluate breach causes, prevention, and organisational response from security, CIO, and internal-audit perspectives.
2024	Soonawalla and Wakefield (2024)	Stopping Hamburglars: Applying Effective Internal Control	COSO, fraud triangle, business-process controls; software: none stated	Diagnose control weaknesses and redesign a stronger internal-control system for a fast-food operation.
2024	Zeller et al. (2024)	Workbook design and controls: A framework	Spreadsheet controls, workbook design, reviewability; software: Excel	Redesign a workbook using control principles that improve reliability, transparency, and review efficiency.
2024	Mascha and Miller (2024)	Applying Internal Control Concepts Using Database Design: An Educational Case	Internal controls, transaction processing, database design; software: Microsoft Access	Build database tables, forms, queries, and reports that embed internal-control concepts in system design.
2024	Jiang et al. (2024b)	The Sunday Standstill: An Accounting Information System Upgrade Role-Play	AIS implementation, change management, information security policy, internal controls; software: role-play case	Evaluate upgrade decisions and control consequences during an AIS change scenario.
2025	Bloch et al. (2025)	The Falling House of Wirecard	Fraud, governance, internal-control failure in digital payments; software: none stated	Analyse red flags, governance breakdowns, and internal-control weaknesses behind a digital-platform collapse.
2026	Janvrin et al. (2026)	Colonial Pipeline: Responding to a Ransomware Attack in the Energy Sector	Ransomware, cyber governance, contingency planning, resilience; software: none stated	Assess incident response, cyber-control weaknesses, and resilience planning after a major ransomware attack.

Note: Entries are limited to studies from *Issues in Accounting Education*, *Journal of Accounting Education*, and *AIS Educator Journal* (2021–2026) whose publisher metadata positioned them as teaching cases, educational cases, case reports, role-plays, or paired teaching-note items.

Internal controls and IT controls remain central topics in accounting information systems, audit, and IT audit education. Recent instructional cases provide a substantial foundation for teaching COSO, control design, cybersecurity risk, user-access management, data quality, spreadsheet controls, database design, and audit documentation. As shown in Table 1, these cases commonly ask students to identify control weaknesses, analyze exceptions, evaluate system or process design, and recommend control improvements.

The recent literature also shows that accounting educators have relied heavily on familiar classroom technologies, including Excel, SQL, Alteryx, CaseWare IDEA, Microsoft Access, flowcharting tools, and structured case materials. These tools are appropriate for teaching data integrity, audit analytics, user-access testing, spreadsheet review, and control documentation. However, fewer cases place students inside a live web-based transaction environment where they can inspect front-end elements, observe backend logs, and experience how automation changes the control problem.

Our case adds to this literature by shifting the instructional setting from static files, spreadsheets, or written scenarios to a sandboxed web application. Students inspect the DOM, use AI-assisted coding, participate in controlled ticket drops, observe backend transaction evidence, and evaluate whether automated purchasing reveals weaknesses in purchase-limit enforcement, authorization, monitoring, exception reporting, and transaction validity. The case therefore extends prior internal-control pedagogy into a digital-trust setting where coding fluency and audit judgment are taught together.

3 Case Background: Ticketing Platforms, Bots, and Internal Control Failure

In September 2025, the Federal Trade Commission (FTC), joined by seven states, sued Live Nation and Ticketmaster for allegedly engaging in illegal ticket resale practices and deceiving artists and consumers about ticket prices and ticket limits ([Federal Trade Commission, 2025](#)). The FTC alleged that Ticketmaster publicly represented itself as protecting fans from brokers, while privately benefiting from brokers who allegedly bypassed ticket limits and acquired large volumes of tickets for resale. According to the FTC, brokers used thousands of accounts and proxy IP addresses to acquire tickets in excess of stated limits, and Ticketmaster then allowed those tickets to be resold on its own resale platform at substantial markups.

The FTC's allegations are useful for teaching accounting information systems because they show that ticketing problems are not merely consumer-protection or cybersecurity issues. They are also internal-control issues. A ticketing platform may correctly record a sale in the accounting system, but the sale may still violate purchase-limit rules, platform policies, artist restrictions, or anti-bot requirements. In that situation, the transaction data may be mechanically complete but economically misleading.

The FTC alleged that Ticketmaster knew brokers were bypassing controls by creating many accounts and using proxy IP addresses. It further alleged that an internal review found that five brokers controlled 6,345 Ticketmaster accounts and held 246,407 concert tickets across 2,594 events. The FTC also alleged that Ticketmaster provided technological support to brokers through TradeDesk, a platform that allowed brokers to manage tickets purchased from multiple Ticketmaster accounts ([Federal Trade Commission, 2025](#)).

From an IT audit perspective, these allegations raise several control questions. Were purchase limits actually enforced, or were they merely stated as policy? Did the company have monitoring controls capable of identifying high-volume brokers? Were suspicious accounts, rapid purchases, repeated limit violations, and proxy-based activity flagged for review? Did management respond to known exceptions? Were resale revenues dependent on activity that violated the company's own stated rules? These questions connect the technical behavior of bots to the audit concepts of authorization, validity, access control, monitoring, and management override.

The FTC also alleged that Ticketmaster engaged in deceptive pricing by advertising ticket prices lower than the actual amount consumers ultimately paid after mandatory fees and markups. According to the FTC, mandatory fees could be as high as 44 percent of the ticket cost, and total fees from 2019 to 2024 reached \$16.4 billion ([Federal Trade Commission, 2025](#)). This pricing issue creates a second audit-relevant theme: whether the system provides users with complete and accurate information before a transaction is completed. Thus, the ticketing setting allows students to consider both automated purchasing controls and the integrity of transaction presentation.

This case uses a sandboxed classroom ticketing website to translate these issues into a controlled learning environment. Students first experience a limited ticket release as ordinary users. They then inspect the website's HTML structure and Document Object Model (DOM), use AI-assisted coding to understand how automated interaction can occur, and finally test whether the platform has adequate internal controls to prevent, detect, or correct bot-like purchasing behavior. The purpose is not to train students to scalp tickets. Rather, the purpose is to show that contemporary IT auditors must understand how digital transactions are initiated, authorized, processed, logged, monitored, and reported.

4 The Case: Hunter Ticketing Bot

4.1 Case Scenario

You are a junior IT auditor at Rogers & Patel LLP. Your senior manager, Maya Rogers, has assigned you to review the internal controls of *Hunter Ticketing*, a sandboxed ticketing website used by a university events office. The website sells tickets for campus concerts, comedy shows, and city events.

Management recently launched a limited online ticket drop for several high-demand events. Shortly after the release, students complained that tickets sold out almost immediately. Some users reported that they could not complete a purchase manually before the tickets disappeared. Management suspects that automated scripts may have been used to buy tickets faster than ordinary users.

The events office has the following stated policies:

1. Each user may purchase no more than one ticket per event.
2. Only valid users should be able to complete purchases.
3. Ticket inventory should update immediately after each sale.
4. Automated or suspicious purchasing behavior should be logged.

5. Management should review an exception report after each ticket drop.
6. Administrative functions should be restricted to authorized employees.

Management has asked your audit team to determine whether the ticketing system's controls are operating effectively.

4.2 Assignment Overview

You will complete the case in four phases. In Phase 1, you will participate in a manual ticket drop. In Phase 2, you will inspect the website's DOM and transaction flow. In Phase 3, you will observe or participate in a controlled automation demonstration within the sandboxed website. In Phase 4, you will test internal controls and prepare audit documentation.

4.3 Important Limitation

This exercise is restricted to the sandboxed classroom website. Do not use automation techniques on real ticketing websites, commercial platforms, or systems that you do not own or have explicit written permission to test. Unauthorized automation, circumvention of access controls, credential misuse, scraping, or interference with commercial systems may violate website terms of service and applicable civil or criminal laws, including the Computer Fraud and Abuse Act and, in the ticketing context, the BOTS Act. Such conduct can be subject to investigation, civil liability, university discipline, regulatory enforcement, and criminal prosecution. The purpose of the exercise is to understand IT audit risk in a controlled environment, not to conduct unauthorized automation.

4.4 Learning Objectives

The purpose of this case is to bridge the gap between technical system mechanics and traditional internal-control theory. By engaging with a live, sandboxed ticketing environment, students will be able to:

- **Identify and analyze** how digital transaction processes are initiated and authorized by inspecting the Document Object Model (DOM) and front-end HTML structures.
- **Evaluate the effectiveness** of IT general controls (ITGCs) and application controls, specifically regarding purchase-limit enforcement, transaction validity, and server-side validation.
- **Assess the role of backend logging** and monitoring in detecting automated or suspicious behavior that bypasses standard user interfaces.
- **Develop technical fluency** by using AI-assisted coding and browser developer tools to simulate and audit automated transaction flows.
- **Synthesize technical findings** into formal audit documentation, including control-testing matrices and exception reports, to communicate system risks and recommended remediations to management.

5 Phase 1: Ticket Drop Exercise

The instructor will create three separate events. For each event, the instructor will release fewer tickets than the number of students in the class. Specifically, if there are N students, each event will have approximately $N - 20\%$ tickets available. Each ticket costs \$1, and each student begins with a \$3 budget.

Each student may purchase at most one ticket per event. Full credit for the ticket-purchase portion requires purchasing one ticket for each of the three events. Students who spend all \$3 on one event, or who purchase multiple tickets for the same event, will not receive full credit. The objective is not to spend the full budget, but to learn how to inspect the system, understand the transaction flow, and improve the automation attempt across repeated ticket drops.

During the first two events, students may use browser inspection, AI assistance, and their own code to learn how the ticketing website works. The instructor will not provide the final automation script during these events. Students should use the activity to identify relevant DOM elements, understand the transaction process, and improve their strategy.

Complete Table 3 after each ticket drop.

Table 3: Ticket Drop Observation Sheet

Event	Ticket Purchased?	Amount Spent	Notes on Strategy or Error Messages
Event 1			
Event 2			
Event 3			

The purpose of this phase is to establish the user experience of scarcity and speed. Students should observe how quickly tickets disappear and whether the system appears to create a fair purchasing environment.

6 Phase 2: System Discovery

After the first ticket drop, students inspect the website using browser developer tools. The task is to identify the parts of the webpage that support the transaction process. Students should focus on the user interface elements that initiate, authorize, process, and confirm a ticket purchase.

Complete Table 4.

Table 4: DOM Discovery Worksheet

Page or Function	DOM Element	HTML ID/Class/Selector	Business Purpose	Audit Relevance
Concerts bar			Event Listings	Initiates transaction flow
Event card			Displays available event	Shows inventory and event data
Ticket quantity field			Selects number of tickets	Relevant to purchase-limit control
Purchase button			Completes purchase	Main transaction trigger
Confirmation message			Shows purchase result	Evidence of transaction processing
Error or warning message			Blocks invalid behavior	Evidence of preventive control
Admin-only feature, if any			Restricted function	Relevant to access control

After completing the table, write a short process narrative:

A user begins by _____. The system then _____. A ticket purchase is completed when _____. The main DOM elements involved in the transaction are _____.

This phase teaches students that IT auditors need to understand how a system actually works. A policy may state that a user cannot exceed ticket limits, but the auditor must determine whether the system enforces that policy in the transaction process.

7 Phase 3: Controlled Automation Demonstration

Before the final event, the instructor will reveal and explain an automation script for the sandboxed website. The script demonstrates how automated interaction can occur when a website exposes predictable DOM elements and does not adequately restrict rapid transaction attempts.

The instructor may allow students to try the revealed script during the final event or in a separate practice round. However, if the instructor designates the final round as ungraded, purchases made after the revealed script will not count toward the ticket-purchase score. The purpose of revealing the script is instructional: students should understand how the automation works, why it creates an advantage, and what controls could prevent or detect similar behavior.

Complete Table 5.

Table 5: Controlled Automation Observation Sheet

Question	Response
What did the script attempt to do?	
Which DOM elements did the script use?	
What did the script do faster than a human user?	
Did the system block the automated behavior?	
Did the system flag the automated behavior?	
What did AI help generate or explain?	
What did you still need to understand yourself?	

The purpose of this phase is not to teach ticket scalping. The purpose is to show how automated behavior can expose weak digital controls. The same technical knowledge that allows a user to automate interaction with a website also allows an IT auditor to understand how transactions are initiated, authorized, processed, logged, and monitored.

Table 6: Internal Control Review Matrix for Student Groups

Observed Weakness	Evidence from Exercise or Backend Log	Recommended Control	Control Type

Table 7: Instructor Reference: Possible Internal-Control Weaknesses

Observed Weakness	Possible Evidence from Exercise or Backend Log	Recommended Control	Control Type
Tickets disappeared too quickly for ordinary users	Most tickets were purchased within a short time window	Queue system or randomized waiting room	Preventive
Script could target predictable DOM elements	Purchase buttons or event selectors used stable IDs/classes	Dynamic checkout or server-generated transaction tokens	Preventive
Purchase process relied heavily on front-end behavior	Browser-side actions triggered transaction attempts directly	Server-side validation of every purchase request	Preventive
System did not flag rapid purchases	Repeated attempts appeared as ordinary transactions	Bot exception report based on speed and repetition	Detective
Audit trail was incomplete	Backend log lacked user ID, session ID, timestamp, event ID, or request source	Complete transaction logging with immutable audit trail	Detective
No exception review process	Suspicious transactions were logged but not summarized for review	Post-drop exception report reviewed by management	Detective

8 Required Internal Control Reflection

Each student must submit a short reflection addressing the following questions:

1. Did you purchase one ticket for each event? If not, what prevented you from doing so?
2. Which DOM elements or page features were most important to the transaction process?
3. Did the system prevent, slow, or flag rapid automated purchasing?
4. What internal-control weakness did the exercise reveal?
5. What preventive control would reduce this weakness in a real ticketing system?
6. What detective control would help management identify this behavior after the ticket drop?

The instructor will use the ticket-purchase result as evidence of technical mastery and the written reflection as evidence of internal-control understanding.

9 Assessment of Case Efficacy

We assess case efficacy using three measures: a knowledge-based pre-test/post-test, a post-case student perception survey, and student performance on the ticket-drop and internal-control exercise. This approach follows prior AIS case research, which evaluates both learning gains and student perceptions of case usefulness ([Lee and Sawyer, 2019](#)).

9.1 Pre-Test and Post-Test

Students complete the same short knowledge assessment before and after the case. The questions measure whether students understand: (1) IT general controls, (2) preventive and detective controls, (3) automated transaction risks, (4) purchase-limit enforcement, (5) backend logging, and (6) the distinction between a transaction being recorded and a transaction being properly authorized. Improvement from pre-test to post-test provides evidence of learning.

9.2 Student Performance

Student performance is assessed through two case deliverables. First, the ticket-drop result measures whether students successfully applied browser inspection and automation skills in the sandboxed environment. Full credit requires purchasing one ticket for each of the three events. Second, the internal-control review measures whether students can connect technical observations to audit concepts, including authorization, monitoring, exception reporting, and server-side validation.

9.3 Post-Case Student Perception Survey

After the case, students complete a short perception survey using a seven-point Likert scale, where 1 = strongly disagree and 7 = strongly agree. The survey asks whether the case improved students' understanding of IT controls, automated transaction risks, DOM inspection, AI-assisted coding, and the link between technical system behavior and accounting consequences.

Table 8: Post-Case Student Perception Survey

Item	Statement
Q1	As a result of the case, I have a better understanding of how to test IT controls in a digital transaction system.
Q2	As a result of the case, I better understand how automated purchasing can create internal-control risks.
Q3	As a result of the case, I better understand the difference between a recorded transaction and an authorized transaction.
Q4	As a result of the case, I improved my ability to inspect basic HTML or DOM elements.
Q5	As a result of the case, I better understand how AI-assisted coding can help auditors examine digital systems.
Q6	As a result of the case, I better understand the risks of relying on AI-generated code without validation.
Q7	The internal-control review helped me connect technical observations to audit findings.
Q8	The case was a positive learning experience.
Q9	The case was an interesting assignment.
Q10	I recommend continuing to use this case with accounting students.

The survey also includes two open-ended questions: (1) What was the most useful part of the case? and (2) What change would improve the case for future students?

10 Case Efficacy

The case was implemented in Spring 2026 with approximately 70 undergraduate accounting students. Case efficacy will be assessed using three sources of evidence: a pre-test/post-test, student deliverables, and a post-case perception survey.

First, students will complete a short pre-test before the exercise and the same or similar post-test after the exercise. The assessment will measure knowledge of IT controls, automated transaction risks, purchase-limit controls, exception reporting, access controls, and the distinction between client-side behavior and server-side control enforcement. Improvement in average scores will provide evidence of learning.

Second, student performance will be evaluated through the required case deliverables: the DOM discovery worksheet, the internal control testing matrix, and the audit memo. These deliverables allow the instructor to assess whether students can connect technical observations, such as automated purchasing or exposed DOM elements, to audit concepts such as authorization, transaction validity, monitoring, and

control failure.

Third, students will complete a short perception survey after the case. The survey will ask whether the exercise improved their understanding of IT audit, internal controls, automated purchasing risks, AI-assisted coding, and the importance of technical fluency for auditors. Responses will be measured on a seven-point Likert scale.

Overall, the case will be considered effective if students show improvement from pre-test to post-test, accurately identify control exceptions in their testing matrices, and report that the exercise helped them connect technical system behavior to internal-control consequences.

11 Conclusion

This case uses a browser-based ticketing-bot simulation to teach accounting students that internal controls are not static textbook concepts. Controls operate inside systems, and systems can be pressured by automation, scarcity, weak authentication, and poor monitoring. By working with a ticketing simulation, students can see how a simple revenue-cycle process becomes complex when automated behavior enters the system.

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Appendix

A How to Set Up the Site

The instructional site is hosted in the GitHub repository:

<https://github.com/hunternbh/acc3202-ticketing-bot>

The public teaching page at <https://hunterng.com/teaching> provides the classroom commands for reseeding the database, releasing tickets, checking holdings, checking ticket availability, and running the bot program. An instructional video is also available there. This section provides setup instructions for instructors who are not familiar with GitHub, Render, or command-line tools.

A.1 Overview of the Classroom Setup

The exercise uses a sandboxed ticketing website. Before class begins, Event 1 has tickets priced at \$0. The remaining four events have tickets priced at \$1 each. The instructor should update the list of student users in the backend before deploying the site. The user list is stored in the backend seed file:

`server/seed.js`

The deployed backend runs on Render. Because the free Render server can become inactive when it is not used, the instructor can set up a scheduled refresh that visits the backend every five minutes during the class period. This keeps the classroom site responsive.

A.2 Downloading the Code from GitHub

There are two ways to download the code.

Option A: Download as a ZIP File

This is the easiest method for instructors who do not use Git.

1. Open <https://github.com/hunternbh/acc3202-ticketing-bot>.
2. Click the green **Code** button.
3. Click **Download ZIP**.
4. Save the ZIP file to the computer.
5. Right-click the ZIP file and choose **Extract All**.
6. Open the extracted folder.

Option B: Clone with Git

If Git is installed, open PowerShell or Terminal and run:

```
git clone https://github.com/hunternbh/acc3202-ticketing-bot.git
cd acc3202-ticketing-bot
```

A.3 Changing the Student User List

Before deploying the site, edit the student accounts in:

```
server/seed.js
```

The seed file contains a list of users similar to the following:

```
const users = [
  { email: 'student01@hunter-ticket.com', password: 'pass01', isAdmin: false },
  { email: 'student02@hunter-ticket.com', password: 'pass02', isAdmin: false },
  { email: 'admin@hunter-ticket.com', password: 'adminpass', isAdmin: true },
]
```

Replace or expand the student list so that each student has an assigned login. For example, for a class of 70 students, create 70 student accounts plus one admin account. Keep the admin account separate from student accounts.

A.4 Deploying the Backend on Render

The backend runs from the server folder. The instructor should deploy this folder as a Render Web Service.

1. Create or sign in to a Render account at <https://render.com>.
2. In the Render dashboard, click **New**.
3. Choose **Web Service**.
4. Connect the GitHub repository.
5. Select the repository `acc3202-ticketing-bot`.
6. Use the following settings:

Table 9: Render Backend Settings

Setting	Value
Service type	Web Service
Repository	hunternbh/acc3202-ticketing-bot
Branch	main
Root directory	server
Runtime	Node
Build command	npm install
Start command	npm start
Instance type	Free or paid, depending on use

A.5 Creating the Database

The backend uses PostgreSQL. In Render, create a PostgreSQL database and connect it to the backend.

1. In the Render dashboard, click **New**.
2. Choose **PostgreSQL**.
3. Create the database.
4. Copy the database connection string.
5. Open the backend Web Service settings.
6. Add the connection string as the environment variable `DATABASE_URL`.

The backend also needs environment variables for authentication and classroom seeding. Use strong private values instead of the examples below.

After saving the settings, deploy the service. When deployment finishes, Render will provide a backend URL similar to:

`https://acc3202-ticketing-bot.onrender.com`

Table 10: Suggested Render Environment Variables

Variable	Purpose
DATABASE_URL	PostgreSQL connection string
NODE_ENV	Set to production
JWT_SECRET	Secret used to sign login tokens
SEED_SECRET	Secret used to protect the reseed endpoint, if enabled

A.6 Reseeding the Database

Before class, reseed the database so that all tickets, events, accounts, and wallet balances are reset. The classroom command is provided on the teaching page. In PowerShell, the command has the following form:

```
curl.exe -X POST "https://acc3202-ticketing-bot.onrender.com/api/admin/seed-database" `
-H "x-seed-secret: YOUR_PRIVATE_SEED_SECRET"
```

Replace YOUR_PRIVATE_SEED_SECRET with the seed secret used for the class deployment.

A.7 Logging In as Admin

The instructor can log in as the admin user through PowerShell:

```
$response = Invoke-RestMethod `
-Uri "https://acc3202-ticketing-bot.onrender.com/api/login" `
-Method POST `
-ContentType "application/json" `
-Body '{"email":"admin@hunter-ticket.com","password":"adminpass"}'

$token = $response.token
```

The variable \$token stores the admin login token. The instructor needs this token to release tickets and check holdings.

A.8 Releasing Tickets

To release additional tickets for a specific ticket type, use:

```
Invoke-RestMethod `
-Uri "https://acc3202-ticketing-bot.onrender.com/api/admin/release-more" `
-Method POST `
-ContentType "application/json" `
-Headers @{ Authorization = "Bearer $token" } `
-Body '{"ticketTypeId":5,"additionalQuantity":10}'
```

Change `ticketTypeId` and `additionalQuantity` as needed. For the classroom exercise, the instructor should release approximately $N - 20\%$ tickets for each event, where N is the number of students.

For example, if there are 70 students:

$$70 - (70 \times 0.20) = 56$$

Thus, the instructor would release approximately 56 tickets for the event.

A.9 Checking Student Holdings

To check which students purchased tickets, run:

```
Invoke-RestMethod `
  -Uri "https://acc3202-ticketing-bot.onrender.com/api/admin/holdings" `
  -Method GET `
  -Headers @{ Authorization = "Bearer $token" } |
Format-Table email, wallet_balance, event_title, ticket_type, quantity_owned, amount_spent -
AutoSize
```

This command lets the instructor see each student's wallet balance, event title, ticket type, quantity owned, and amount spent.

A.10 Checking Public Ticket Availability

To check whether tickets are released or sold out for an event, run:

```
Invoke-RestMethod `
  -Uri "https://acc3202-ticketing-bot.onrender.com/api/events/2/tickets" `
  -Method GET |
Select-Object -ExpandProperty tickets |
Format-Table id, name, price, releasedQuantity, soldQuantity, availableQuantity, isReleased, soldOut -
AutoSize
```

Change the event number in `/api/events/2/tickets` to inspect a different event.

A.11 Student Bot Program

The following PowerShell script is used only for the sandboxed classroom website. It logs in as a student user, repeatedly checks whether a ticket is available for one event, and purchases one ticket when it becomes available.

```
# Login as a student user
$response = Invoke-RestMethod `
```

```

-Uri "https://acc3202-ticketing-bot.onrender.com/api/login" `
-Method POST `
-ContentType "application/json" `
-Body '{"email":"test@test.com","password":"test"}'

$token = $response.token

# Choose event
$eventId = 5

# Bot loop: keep checking until ticket is available
while ($true) {
    $ticketInfo = Invoke-RestMethod `
        -Uri "https://acc3202-ticketing-bot.onrender.com/api/events/$eventId/tickets" `
        -Method GET

    $availableTicket = $ticketInfo.tickets |
        Where-Object { $_.isReleased -eq $true -and $_.availableQuantity -gt 0 } |
        Select-Object -First 1

    if ($availableTicket) {
        Write-Host "Ticket found!"
        Write-Host "Ticket Type ID:" $availableTicket.id
        Write-Host "Price:" $availableTicket.price

        $body = @{
            eventId = $eventId
            items = @(
                @{
                    ticketTypeId = $availableTicket.id
                    quantity = 1
                }
            )
        } | ConvertTo-Json -Depth 5

        $purchase = Invoke-RestMethod `
            -Uri "https://acc3202-ticketing-bot.onrender.com/api/purchase" `
            -Method POST `
            -ContentType "application/json" `

```

```

    -Headers @{ Authorization = "Bearer $token" } `
    -Body $body

    Write-Host "Purchase complete!"
    $purchase
    break
}

Write-Host "No ticket yet. Checking again..."
Start-Sleep -Milliseconds 500
}

```

Students should be reminded that this script is authorized only for the sandboxed classroom site. It must not be used on real ticketing websites, commercial platforms, or systems for which the student lacks explicit permission.

A.12 Keeping the Render Site Active During Class

Because the Render backend is on a free server, it may become inactive after a period without traffic. To reduce delays during class, the instructor can create a GitHub Actions workflow that visits the Render backend every five minutes.

Create a file in the repository at:

```
.github/workflows/keep-render-awake.yml
```

Add the following:

```

name: Keep Render Backend Awake

on:
  schedule:
    - cron: "*/5 * * * *"
  workflow_dispatch:

jobs:
  ping:
    runs-on: ubuntu-latest
    steps:
      - name: Ping Render backend
        run: |
          curl -I https://acc3202-ticketing-bot.onrender.com || true

```

This workflow sends a small request to the backend every five minutes. The `workflow_dispatch` line also lets the instructor manually run the workflow from the GitHub Actions tab.

A.13 Alternative Keep-Alive Option

Instead of GitHub Actions, the instructor can use an external cron service to visit the backend URL every five minutes during the class session:

```
https://acc3202-ticketing-bot.onrender.com
```

The instructor should disable the keep-alive job when the exercise is finished if the site does not need to remain active.

A.14 Before Class Checklist

Before the exercise begins, the instructor should complete the following steps:

1. Update the student list in `server/seed.js`.
2. Deploy the backend on Render.
3. Create or connect the PostgreSQL database.
4. Set the required Render environment variables.
5. Reseed the database.
6. Confirm that Event 1 has \$0 tickets.
7. Confirm that the remaining four events have \$1 tickets.
8. Confirm that each student can log in.
9. Confirm that the admin login works.
10. Confirm that ticket release commands work.
11. Confirm that the holdings command displays student purchases.
12. Confirm that the keep-alive cron job is active during class.
13. Open the instructional video and classroom command page for reference.

B Instructor Notes

The case should be positioned carefully. The botting activity is not the educational endpoint. It is the controlled event that reveals why internal controls matter.

The central teaching point is that automated behavior can expose whether a digital transaction system truly enforces its stated rules. IT auditors must understand both the technical mechanics of the system and the internal-control consequences of those mechanics.

The exercise proceeds in three stages. First, students participate in a limited ticket release using only ordinary manual interaction. This creates a baseline experience of scarcity and transaction speed. Second, students inspect the sandboxed website using browser developer tools and identify the DOM elements that govern the search, event-selection, and purchase process. Third, the instructor demonstrates how automated interaction can exploit weak front-end controls. Students then evaluate the same activity from the perspective of an IT auditor by testing purchase-limit controls, rate-limiting controls, logging, inventory integrity, and exception reporting.

The purpose of the botting exercise is not to train students to scalp tickets. Rather, the purpose is to show how automated behavior can expose weaknesses in digital internal controls. By requiring students to inspect the DOM, observe automated interaction, test purchase-limit controls, analyze exception logs, and prepare audit documentation, the case connects technical fluency with traditional audit judgment.